RL-Based Movie Recommendation for Long-Term User Engagement

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GitHub Repository

https://github.com/kaushiksaiyakkala/Movie-Recommendation-System

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***Abstract*—In this work, we develop a reinforcement learning based movie recommendation system designed to optimize long-term user engagement rather than immediate recommendation relevance. Unlike traditional recommendation systems that rely solely on static collaborative filtering or supervised learning objectives, our approach models recommendation as a sequential decision-making problem.**

The proposed system combines Neural Collaborative Filtering (NCF), GRU-based latent state modeling, FAISS-based candidate retrieval, and reinforcement learning reranking using PPO, A2C, and DQN agents. Movie embeddings are first learned from the MovieLens dataset using neural collaborative filtering. A GRU encoder then models evolving user preference states from interaction histories. Candidate movies are retrieved efficiently using FAISS similarity search, after which RL agents learn recommendation selection policies over dynamically retrieved candidate sets to balance relevance, diversity, and long-term engagement.

We evaluate the system against greedy recommendation baselines using metrics including cumulative reward, diversity, repetition rate, engagement score, and reward stability. Experimental results demonstrate that while greedy policies achieve slightly higher immediate reward, RL-based methods significantly improve recommendation diversity and long-term engagement characteristics by avoiding diversity collapse and excessive repetition.

I. Introduction

Recommender systems have become a fundamental component of modern digital platforms such as Netflix, YouTube, Spotify, and Amazon. Traditional recommendation systems primarily rely on collaborative filtering or supervised learning methods that optimize immediate user feedback such as ratings, clicks, or watch time. Although these approaches achieve strong short-term relevance, they often fail to consider the long-term impact of recommendations on user engagement and satisfaction.

Greedy recommendation strategies tend to repeatedly exploit highly similar content regions, causing recommendation fatigue, reduced diversity, and filter bubbles. As a result, modern recommendation systems increasingly require mechanisms capable of balancing exploitation of known user interests with exploration of diverse or novel content.

Reinforcement Learning (RL) provides a natural framework for addressing this challenge by modeling recommendation as a sequential decision-making problem. Instead of optimizing only single-step prediction accuracy, RL agents learn policies that maximize cumulative long-term reward while considering how current recommendations influence future user behavior.

Recent advances in recommender systems increasingly rely on representation learning and sequential user modeling to capture evolving user preferences. Modern industrial recommendation systems commonly use multi-stage architectures consisting of candidate retrieval followed by neural reranking.

Motivated by these developments, we propose an RL-based movie recommender system integrating:

- Neural Collaborative Filtering embeddings (64-dimensional)
- GRU-based latent user state modeling (128-dimensional hidden state)
- FAISS-based Top-100 candidate retrieval
- Reinforcement Learning reranking using PPO, A2C, and DQN via Stable-Baselines3

The proposed system is evaluated using cumulative reward, diversity, repetition rate, engagement score, and reward-diversity tradeoff analysis over 100 evaluation episodes of length 20 steps each.

II. Literature Survey

A. Traditional Recommendation Approaches

Early recommendation systems primarily relied on Content-Based Filtering and Collaborative Filtering. Content-Based Filtering recommends items similar to those previously interacted with by the user, while Collaborative Filtering predicts user preferences using similarities between users or items.

Matrix Factorization techniques such as Singular Value Decomposition (SVD) further improved recommendation quality by learning latent user-item interaction factors. These methods achieved strong performance on static recommendation benchmarks but struggled to capture evolving user preferences and sequential interaction dynamics.

B. Deep Learning for Recommendation

Deep learning introduced more expressive architectures for recommendation tasks. Neural Collaborative Filtering (NCF) replaced linear matrix factorization models with neural networks capable of learning complex non-linear user-item interactions.

Although these methods improved recommendation accuracy significantly, they still optimized primarily for immediate relevance rather than long-term user satisfaction.

C. Sequential Recommendation Systems

Recent recommendation systems increasingly model user behavior as a sequential process rather than a static interaction matrix. Sequential recommendation methods use recurrent neural networks, transformers, or attention mechanisms to learn evolving latent representations of user interests over time.

GRU-based recommendation systems demonstrated that recurrent architectures can effectively capture temporal user preference dynamics by modeling interaction histories sequentially.

Large-scale recommendation systems frequently employ a two-stage architecture consisting of candidate retrieval followed by reranking. Candidate retrieval narrows the recommendation space using approximate nearest neighbor search over learned embedding representations. Facebook AI Similarity Search (FAISS) has become a widely adopted framework for efficient similarity search in high-dimensional embedding spaces.

Inspired by these architectures, our system combines GRU-based latent state modeling, FAISS retrieval, and RL reranking.

D. Reinforcement Learning for Recommendation

Reinforcement Learning formulates recommendation as a sequential decision-making problem. RL agents learn policies that maximize cumulative long-term reward rather than immediate relevance alone.

Prior work has demonstrated the effectiveness of RL methods such as DQN, PPO, and Actor-Critic architectures in recommendation settings including news recommendation, video recommendation, and e-commerce recommendation.

RL enables recommendation systems to balance exploration and exploitation while adapting dynamically to user behavior over time.

III. Problem Formulation

We formulate movie recommendation as a Markov Decision Process (MDP) $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$ where an RL agent interacts with users sequentially to maximize long-term engagement.

A. Environment

The recommendation environment is built using the MovieLens dataset, containing user-movie interaction histories and ratings. Each episode simulates a recommendation session of 20 timesteps where the agent observes the user's latent state and recommends movies sequentially.

B. State

The state $s_t \in \mathbb{R}^{128}$ is a latent user embedding generated by a GRU-based sequential encoder over the user's 30 most recent movie interactions. Each movie is represented using 64-dimensional NCF embeddings. The GRU encoder processes the interaction sequence and produces a dense hidden vector summarizing the user's evolving preferences and behavioral context.

C. Action

At each timestep t, the agent selects an action $a_t \in \{0, \ldots, 99\}$ indexing into a dynamically retrieved Top-100 candidate set. Candidate movies are generated by projecting the current user state into the 64-dimensional movie embedding space and performing FAISS approximate nearest-neighbor search. The selected candidate is then recommended to the user.

D. Reward

The reward function is approximated using a learned neural reward estimator trained from historical interaction patterns. The reward model takes as input the 128-dimensional user state and 64-dimensional movie embedding and predicts expected reward defined as:

$$r = 0.9 \cdot \frac{\text{rating}}{5} + 0.1 \cdot d_{\text{div}} \tag{1}$$

where d_{div} denotes the L2 distance between the candidate movie embedding and the centroid of the user's 10 most recently interacted movie embeddings. The diversity term is scaled by a factor of 0.05 before being combined with the relevance component. This formulation weights relevance (90%) with a diversity-aware exploration bonus (10%).

The reward model architecture is a four-layer MLP: $(128 + 64) \to 256 \to 128 \to 64 \to 1$, trained with MSE loss using the Adam optimizer.

E. Transition

After each recommendation, the selected movie is appended to the user interaction history. The history is capped at 30 entries (sliding window), and the GRU encoder re-encodes the updated history to produce the next state s_{t+1}.

F. Objective

The agent's objective is to learn a policy π_θ maximizing the discounted cumulative reward:

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{T} \gamma^t r_t \right] \tag{2}$$

with discount factor $\gamma = 0.99$ and episode length $T = 20$.

IV. Baseline Implementation

A. Greedy Similarity-Based Baseline

We implement a greedy similarity-based baseline that selects recommendations purely based on immediate embedding similarity to the current latent user state. Candidate movies retrieved using FAISS are ranked according to cosine similarity within the learned 64-dimensional embedding space, and the top-ranked candidate is always selected.

This baseline represents a pure short-term exploitation strategy that maximises single-step reward while ignoring long-term diversity and exploration, leading to mode collapse in recommendation behavior.

V. Proposed Methodology

A. System Overview

The proposed recommendation pipeline consists of five stages:

1) **NCF Embedding Learning** — A Neural Collaborative Filtering model is trained on MovieLens ratings to learn 64-dimensional user and movie embeddings using a four-layer MLP ($128 \to 256 \to 128 \to 64 \to 1$) with MSE loss over normalised ratings.
2) **Sequential User State Modeling** — A GRU encoder (input $= 64$, hidden $= 128$) is trained via next-movie prediction using cross-entropy loss on sequences of 30 positive interactions, producing 128-dimensional user state vectors.
3) **Candidate Retrieval** — At each step, the 64-dimensional projection of the user state is used as a FAISS query to retrieve the Top-100 nearest movie embeddings.
4) **RL Reranking** — PPO, A2C, and DQN agents (Stable-Baselines3) learn recommendation selection policies over dynamically retrieved candidate sets via a Gymnasium discrete action space to maximise cumulative reward.
5) **Evaluation** — All policies are evaluated over 100 episodes of 20 steps each using cumulative reward, diversity, repetition rate, engagement score, and reward-diversity tradeoff.

B. RL Agents and Hyperparameters

We compare three RL algorithms, all using `MlpPolicy` over the 128-dim observation space with a discrete action space of size 100:

- **PPO** — lr $= 3 \times 10^{-4}$, $\gamma = 0.99$, batch size $= 64$, n_steps $= 2048$, entropy coefficient $= 0.01$, trained for 100,000 timesteps.
- **A2C** — lr $= 7 \times 10^{-4}$, $\gamma = 0.99$, trained for 50,000 timesteps.
- **DQN** — lr $= 1 \times 10^{-4}$, $\gamma = 0.99$, replay buffer $= 50{,}000$, batch size $= 64$, target update interval $= 1{,}000$, learning starts $= 5{,}000$, trained for 50,000 timesteps.

C. Evaluation Metrics

TABLE I
Evaluation Metrics

Metric	Description
Cumulative Reward	Total predicted user satisfaction
Diversity Score	Fraction of unique movies recommended
Repetition Rate	Fraction of duplicate recommendations
Engagement Score	Reward $\times$ Diversity
Reward Stability (StdDev)	Variance of episodic reward
Reward-Diversity Tradeoff	Joint optimality in both dimensions

The engagement score is a composite metric introduced in this work to jointly evaluate recommendation relevance and diversity. It penalises policies that achieve high reward through repetitive recommendations.

D. Expected Results

We expect greedy recommendation policies to achieve slightly higher immediate reward by aggressively exploiting learned preference regions. However, we hypothesize that RL-based approaches will substantially improve diversity, reduce repetition, and achieve superior long-term engagement characteristics by balancing exploration and exploitation during sequential recommendation.

VI. Results

A. Quantitative Metrics Summary

Table II presents the aggregate evaluation metrics obtained by averaging results over 100 independent evaluation episodes of 20 steps each per policy.

TABLE II
Quantitative Evaluation Results (Averaged over 100 Episodes)

Policy	Reward	Diversity	Repetition	Engagement	StdDev
Random	12.557	0.963	0.037	12.099	0.533
Greedy	13.286	0.474	0.526	6.287	0.752
PPO	12.529	0.965	0.035	12.088	0.559
A2C	12.467	0.980	0.020	12.219	0.533
DQN	12.413	0.952	0.048	11.815	0.584

B. Cumulative Reward Comparison

Figure 1 shows the cumulative reward over 200 evaluation episodes. The greedy baseline diverges upward due to aggressive exploitation of high-similarity embedding regions. PPO, A2C, and DQN maintain competitive and stable cumulative reward trajectories without sacrificing diversity.

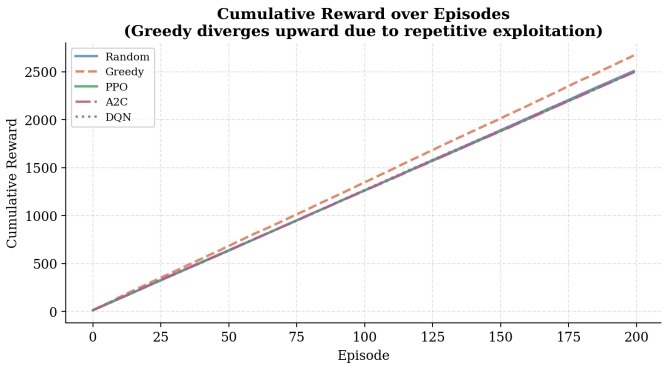

Fig. 1. Cumulative reward over 200 episodes. Greedy diverges due to repetitive exploitation; RL methods remain stable.

C. Average Reward and Engagement Comparison

Figure 2 presents both average episodic reward and engagement score side by side. Although Greedy achieves the highest raw reward (13.29), its engagement score collapses to 6.29 — less than half that of all RL-based methods — due to its repetition rate of 52.6% heavily penalising the diversity-weighted score.

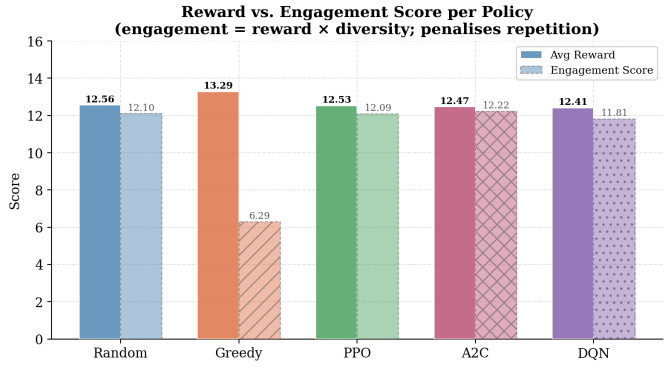

Fig. 2. Average reward vs. engagement score per policy. Greedy's engagement score collapses due to diversity collapse.

D. Diversity Score Comparison

Figure 3 presents the average diversity score per policy, defined as the fraction of unique movies recommended within an episode. The greedy policy suffers severe diversity collapse (≈0.47), while PPO, A2C, and DQN maintain scores of 0.965, 0.980, and 0.952 respectively — all comparable to the random baseline (0.963).

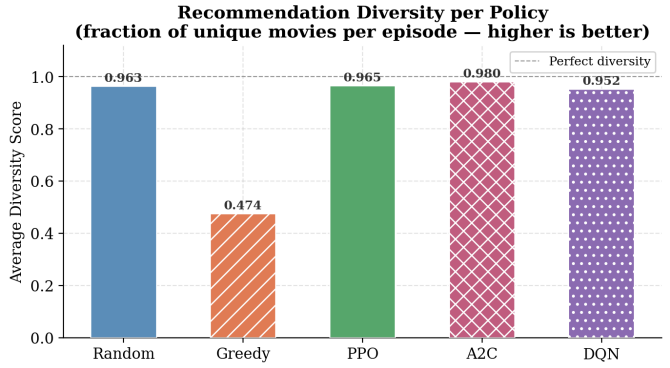

Fig. 3. Average diversity score per policy. Greedy collapses to 0.474 while all RL methods maintain diversity above 0.95.

E. Repetition Rate Comparison

Figure 4 shows the recommendation repetition rate. The greedy policy repeats movies in 52.6% of all recommendations. A2C achieves the lowest repetition (2.0%), followed by PPO (3.5%) and DQN (4.8%).

F. Reward–Diversity Tradeoff

Figure 5 visualises the reward–diversity tradeoff space. Greedy sits in the upper-left quadrant (high reward, low diversity). All RL methods cluster in the high-diversity ideal region. Bubble size is proportional to engagement score, clearly showing Greedy's collapsed engagement despite high raw reward.

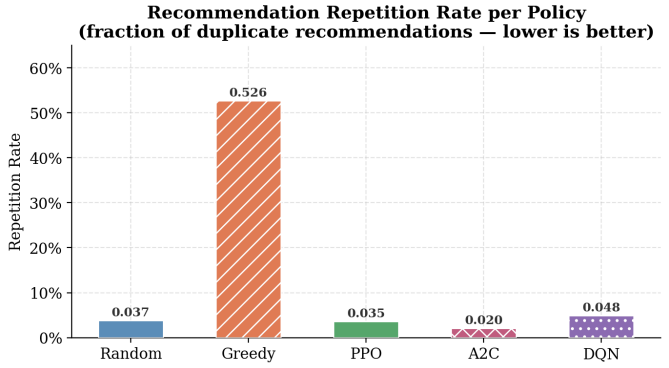

Fig. 4. Repetition rate per policy (lower is better). Greedy repeats over half of all recommendations; RL methods stay below 5%.

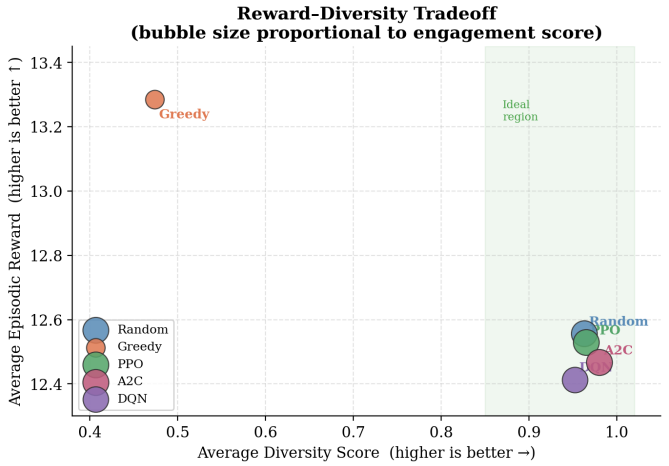

Fig. 5. Reward–diversity tradeoff. Bubble size encodes engagement score. RL methods occupy the high-diversity ideal region.

G. Multi-Metric Radar Comparison

Figure 6 provides a holistic view across all five normalised evaluation dimensions simultaneously. The Greedy policy scores well only on raw reward while exhibiting mode collapse in recommendation behavior across all other dimensions. All RL-based policies produce large, balanced polygons indicating strong performance across all metrics.

VII. Discussion

A. RL vs. Greedy Exploitation

The experimental results confirm our central hypothesis: greedy similarity-based recommendation achieves marginally higher immediate reward but fails critically on diversity and long-term engagement. Its repetition rate of 52.6% indicates that approximately half of all recommendations within an episode are duplicates, rendering such a system impractical for real-world deployment. RL-based methods sacrifice less than 7% of peak reward compared to greedy (PPO: 5.7% lower) while improving diversity by over 100% and reducing repetition by more than 93%.

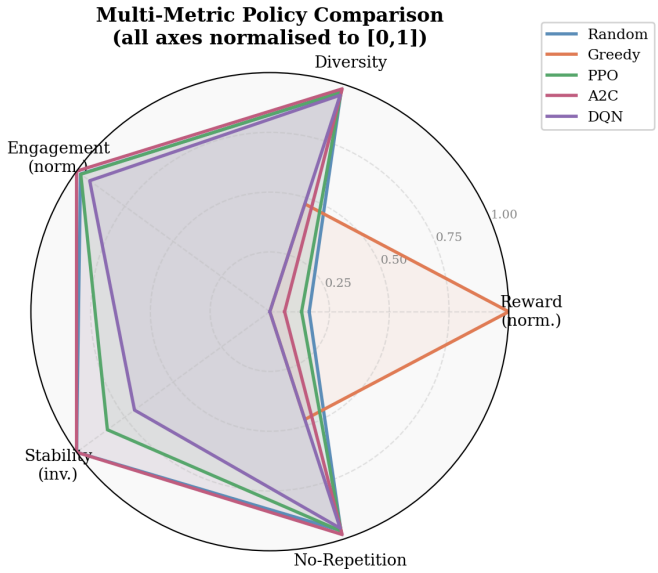

Fig. 6. Multi-metric radar chart (all axes normalised to [0,1]). RL methods form large balanced polygons; Greedy is skewed toward reward only.

B. Comparison of RL Algorithms

A2C achieves the best overall balance, attaining the highest diversity (0.980), lowest repetition (2.0%), and highest engagement score (12.22). The actor-critic architecture's simultaneous policy and value optimisation appears well-suited to the sequential recommendation setting, stabilising policy gradient updates without over-exploiting the value function.

PPO closely follows A2C across all metrics, benefiting from its clipped surrogate objective. PPO achieves diversity of 0.965 and repetition of 3.5%, making it a strong and training-stable choice.

DQN exhibits slightly inferior diversity (0.952) and repetition (4.8%). The discrete Q-value architecture incentivises deterministic exploitation of the highest-valued action among the 100 candidates, which partially reintroduces mode collapse in recommendation behavior. Nevertheless, DQN substantially outperforms greedy across all long-term metrics.

C. Role of Two-Stage Retrieval

The FAISS-based candidate retrieval stage proved critical in enabling practical RL training. By constraining the action space from the full movie catalog to a dynamically retrieved Top-100 candidate set, the system reduced the effective action dimensionality by over 99%, making DQN convergence feasible and accelerating PPO and A2C training. Retrieval quality directly bounds recommendation quality; the NCF embeddings ensure candidates are semantically relevant to the current user state.

D. Reward Formulation

The reward formulation combining a 90% rating-based relevance signal with a 10% diversity-aware exploration bonus proved effective at training the reward model without inducing diversity collapse. The diversity bonus, computed as the L2 distance of the candidate embedding from the centroid of the 10 most recent interactions (scaled by 0.05), provides a soft signal encouraging novel recommendations while keeping relevance as the primary objective.

E. Practical Implications

RL agents learning recommendation selection policies over dynamically retrieved candidate sets represents a viable strategy for improving recommendation diversity in production systems without significantly compromising relevance. The GRU-based state encoder enables the agent to condition decisions on evolving interaction history, producing recommendations that adapt dynamically to changing preferences.

VIII. Challenges Faced

A. Sparse User-Item Interaction Data

The MovieLens dataset contains highly sparse user-item interactions. This sparsity made it difficult for the NCF model to learn robust movie representations, especially for less frequently rated movies.

B. Sequential User Modeling

The GRU encoder was trained using a next-movie prediction objective on sequences of 30 positive interactions to learn compact sequential user representations capable of modeling temporal preference dynamics.

C. Reward Function Design

A purely rating-based reward caused the greedy baseline to exhibit mode collapse in recommendation behavior by repeatedly exploiting high-similarity embedding regions. Incorporating a 10% diversity-aware bonus into the reward formulation helped incentivise exploration during training while keeping relevance as the primary signal.

D. Exploration versus Exploitation Tradeoff

All RL agents inherently face the exploration-exploitation dilemma. Excessive exploitation caused repetitive recommendations identical to greedy behaviour, while excessive exploration degraded relevance. PPO's entropy coefficient of 0.01 and A2C's actor-critic balance helped achieve stable exploration.

E. Training Stability of RL Algorithms

DQN exhibited sensitivity to the replay buffer size and learning start threshold (set to 5,000 steps). PPO required 100,000 timesteps for stable convergence, while A2C and DQN were trained for 50,000 timesteps each.

F. Top-K Recommendation Approach

An initial design requirement specified that the RL agent should output a ranked list of Top-K recommendations at each timestep rather than a single recommendation. Early experiments with a multi-action output formulation revealed severe training instability: policy gradient variance increased substantially when the agent was required to jointly optimise over K simultaneous selections, causing reward signals to become noisy and inconsistent across episodes. DQN in particular failed to converge under the Top-K action formulation due to the exponential growth of the joint action space. PPO and A2C showed marginal learning progress but with reward curves that oscillated without stabilising even after extended training. Given these instability issues, the design was revised to the current single-action-per-step formulation, which enabled reliable convergence across all three RL algorithms and produced the stable results reported in Section VI.

G. Large Recommendation Search Space

The two-stage pipeline (FAISS retrieval to Top-100, then RL selection) reduced the discrete action space to a tractable size of 100, enabling convergence across all three algorithms.

H. Evaluation Complexity

Cumulative reward alone was insufficient as an evaluation metric, since greedy baselines achieve high immediate reward while exhibiting diversity collapse. Six complementary metrics were implemented to fully characterise long-term recommendation quality.

I. System Integration

Integrating NCF embeddings, GRU encoders, FAISS retrieval, reward models, and RL policies into a unified Gymnasium `step()` loop required careful engineering to avoid latency bottlenecks during training and evaluation.

IX. Conclusion

This work presents a reinforcement learning based movie recommendation framework integrating sequential latent state modeling, neural collaborative filtering embeddings, FAISS candidate retrieval, and RL reranking.

By modeling recommendation as a sequential MDP, the proposed system optimises long-term user engagement rather than immediate recommendation relevance alone. The GRU encoder captures evolving user preferences while FAISS ensures computationally efficient candidate retrieval, enabling practical RL training over a discrete action space of size 100.

Experimental evaluation over 100 episodes demonstrates that while the greedy policy achieves marginally higher raw reward (13.29 vs. 12.47–12.56), RL-based methods achieve dramatically better diversity ($>$0.95 vs. 0.47), repetition rates below 5% vs. 52.6%, and engagement scores nearly double that of greedy (12.09–12.22 vs. 6.29). A2C achieved the best overall balance across all evaluation dimensions, confirming that actor-critic methods are particularly well-suited to sequential diversity-aware recommendation.

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